Wildland fire prevention: a **review**

H Hesseln - Current Forestry Reports, 2018 - Springer

… regarding human **behavior** and factors that will lead to enhanced **wildfire** prevention efforts.
… and effective methods to **model** the likelihood of **wildfire** including those resulting from arson. …

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Generative AI as a Pillar for Predicting 2D and 3D **Wildfire Spread**: Beyond Physics-Based Models and Traditional Deep Learning

H Xu, S Zlatanova, R Liang, I Canbulat - Fire, 2025 - mdpi.com

… AI in **wildfire management**, including unified multimodal modeling of 2D and 3D **dynamics**, …
review of studies applying generative AI to **wildfire prediction**, highlighting their strengths for …

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A method for ensemble **wildland fire simulation**

MA Finney, IC Grenfell, CW McHugh, RC Seli… - … Modeling & Assessment, 2011 - Springer

… An ensemble **simulation** system that accounts for uncertainty in long-range weather conditions
and two-dimensional **wildland fire spread** is described. Fuel moisture is expressed based …

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BA Ebel, ZM Shephard, MA Walvoord… - Earth's …, 2023 - Wiley Online Library

… Building a community **wildfire model** that simulates the links between **wildfire spread**, **wildfire**
effects on vegetation and soils, hydrologic and erosion processes, biogeochemical cycling, …

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Review and perspectives of digital twin systems for **wildland fire management**

Y Li, T Zhang, Y Ding, R Wadhwani… - Journal of Forestry …, 2024 - Springer

… This **review** examines the potential utility of Digital Twin in **wildfire management** and aims
to inspire … It functions as a low-cost and fast operation **model** to **model** the **wildfire spread** by …

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Review of **wildfire** modeling considering effects on land surfaces

D Or, E Furtak-Cole, M Berli, R Shillito… - Earth-Science …, 2023 - Elsevier

… **review** are: (1) to introduce hydrologists, soil scientists and ecologists to principles and advances
in modeling of **wildfire dynamics** and rates of **spread** … large eddy **simulation** models. We …

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Forest fire management and fire suppression strategies: a systematic literature **review**

B Tezcan, T Eren - Natural Hazards, 2025 - Springer

… plays a vital role in controlling and preventing the **spread** of fires. The combination of … This
study provides a comprehensive **review** of mathematical models for **forest fire management** …

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Review of vortices in **wildland fire**

JM Forthofer, SL Goodrick - Journal of Combustion, 2011 - Wiley Online Library

… fire **behavior** and safety concerns. In this paper, the current state of knowledge of the interaction
of **wildland fire** … In this study a computer **model** of the atmospheric boundary layer was …

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A cellular automaton **model** of **wildfire propagation** and extinction

KC Clarke, JA Brass, PJ Riggan - … Engineering and Remote …, 1994 - research.fs.usda.gov

… We propose a new **model** to predict the spatial and temporal **behavior** of wildfires. Fire **spread** and intensity were simulated using a cellular automaton **model**. Monte Carlo techniques …

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SPH Boroujeni, A Razi, S Khoshdel, F Afghah… - Information …, 2024 - Elsevier

… for **wildfire** classification, segmentation, detection, and monitoring tasks. Ultimately, we underscore the substantial advancement in **wildfire** … to understand dynamic **wildfire behavior**. …

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